

Problem Set 8

E. Count GCD

tianzheyu submissions

#	When	Who	Problem	Lang	Verdict	Time	Memory
385718577	Aug/05/2026 15:27UTC+10	tianzheyu	D - Count GCD	C++23 (GCC 14-64, msys2)	Accepted	359 ms	100 KB

Process

I used the prefix GCD property $\gcd(a_{i-1}, b_i) = a_i$ to simplify the problem into finding how many numbers $Y \leq m / a_i$ are coprime to $X = a_{i-1} / a_i$. To count these efficiently, I extracted the prime factors of X and applied the Inclusion-Exclusion Principle using bitmasking. I then multiplied the valid counts for each position modulo 998244353.

Challenges and Overcoming

I ran into a bug where my program would output completely wrong answers or desynchronize on later test cases. I found it when I used a `break` statement immediately after detecting an impossible sequence ($a_{i-1} \not\equiv 0 \pmod{a_i}$). Because I was reading the array elements dynamically, breaking early left unread numbers in the `cin` stream, which were then incorrectly read as n and m for the next test case. Next time to prevent this bug, I will always read the entire input array into a `vector` first before applying any early-exit or `break` logic, ensuring the input stream remains perfectly synchronized.